CITIZEN SCIENCE & PLANETARY FORENSICS • FEATURE ARTICLE

How AI Lowered the Barrier to Rigorous Science

An Inquisitive Amateur, an AI Pair Programmer, and JPL Space Race Heritage Audit Raw NASA Telemetry on an Anomalous Cylinder in Gale Crater

By Justin Bogner • October 2026 • Technical Report CMR-2026-01 • DOI: 10.5281/zenodo.23789092



The Sol 3556 Anomaly in Context: A small, highly regular cylinder embedded in the coarse regolith of Paraitepuy Pass, Gale Crater. Captured by NASA's *Curiosity* rover using the Mastcam Right 100 mm camera on August 7, 2022 (PDS Product: 3556MR1025170721700585C00).

It started with a picture

In August 2022, NASA's *Curiosity* rover took a photo on Mars. In the lower right corner was something strange. There appeared to be a cylinder poking out of the dirt. It's surrounded by pebbles and dirt, things that one would expect to find on Mars. The object was, at least to me, obviously anomalous.

People noticed. Some said it was debris from the rover. Some said it was just a strange rock. And of course some people immediately jumped to aliens. Even astrophysicist Avi Loeb wrote that NASA should take a closer look.

I'd seen it discussed on podcasts and YouTube videos, but no one ever seemed to actually meaningfully dig into the data. No one seemed to be doing any actual work, just making a lot of noise. This applies both directions, from the scientific community and the more heterodox crowd.

I began this project wanting to determine the actual dimensions and whether or not it was mathematically cylindrical. This seemed relatively straightforward and in my mind would bring some clarity to the discussion.

However, seeing familiar shapes in chaos, and our desire to see what we want to see is a very real, uniquely human trait. I can watch a podcast, see a photo someone presents of a "pyramid" or a face, and just because of who I am I am inclined to believe it. This is likely why the scientific community is skeptical of such claims. So I had to constrain myself to the raw telemetry, the resulting data, and nothing else.

On that note, a disclaimer: I'm not an engineer and I'm not a mathematician. I've never taken a course in computer vision or planetary photogrammetry. However, I am inquisitive and stubborn, probably have too much time on my hands, and happen to have a fun family connection to this work.

A family footnote on Mars

This story begins with another story: one that makes what is to come more interesting. Not more credible, but certainly more entertaining. We begin in Decatur, Indiana, in 1931. My great-grandmother Flossie gave birth to a baby boy, Richard "Sam" Bogner, whose work would, a few decades later, make the work I have done today possible. Named after his father, my great-uncle Sam would eventually graduate from Ball State Teachers College in Muncie, now Ball State University, in 1959. His career as a chemical engineer began with Delco Remy, a major automotive electrical parts manufacturing division of General Motors located in Anderson, Indiana. In 1966, he joined the exodus of young men from this region, an area still known for producing top-tier engineering talent, who took their degrees to the front lines of the space race. Sam relocated his family to Pasadena, California, to work at NASA's Jet Propulsion Laboratory, or JPL.

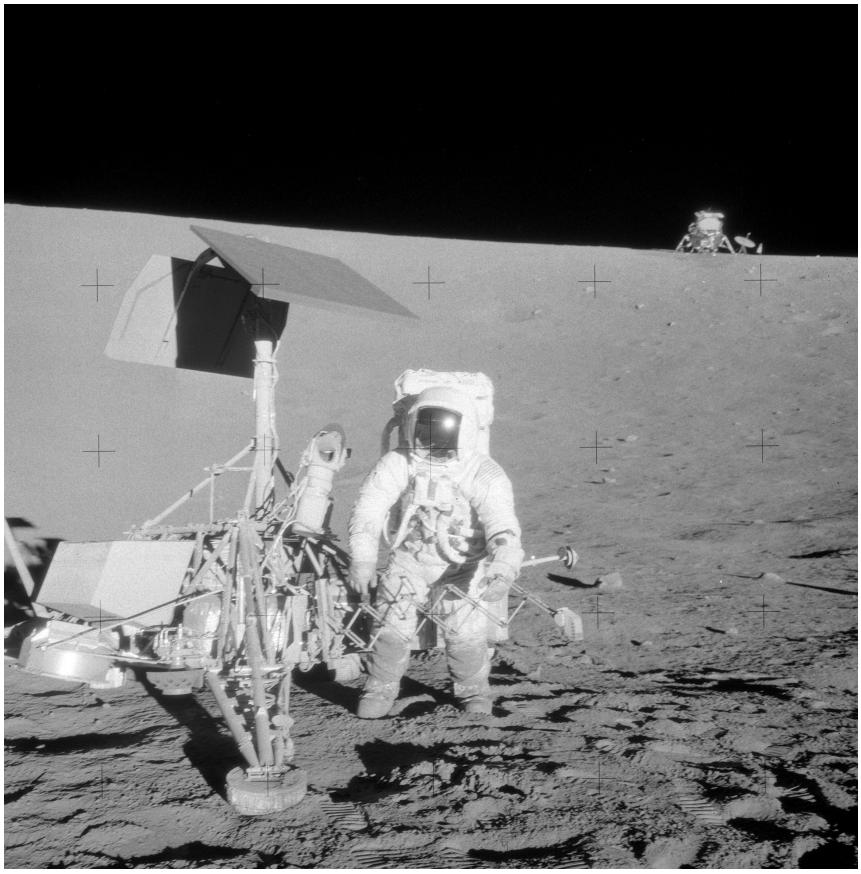
During his time at JPL, he worked on evaluating and perfecting the high-energy silver-zinc battery packs that would power the Surveyor lunar landers. It wasn't just about making the batteries hold a charge; his team had to ensure these batteries could withstand the violent vibrations of a rocket launch, survive the deceleration of a lunar touchdown, and keep the spacecraft alive through freezing, two-week-long lunar nights where temperatures plummeted to hundreds of degrees below zero.

From the Moon, his focus shifted outward to the Red Planet. He became the project engineer and the definitive voice on battery flight performance for the Mariner program, specifically Mariners 6, 7, 9, and 10. I looked up the historic NASA technical reports from the Mariner Mars 1971 mission, and seeing his name right there on the cover was profoundly humbling. His work tracked how deep-space silver-zinc and nickel-cadmium cells degraded over months of interplanetary transit, managed the chemistry of separator systems to prevent short circuits,





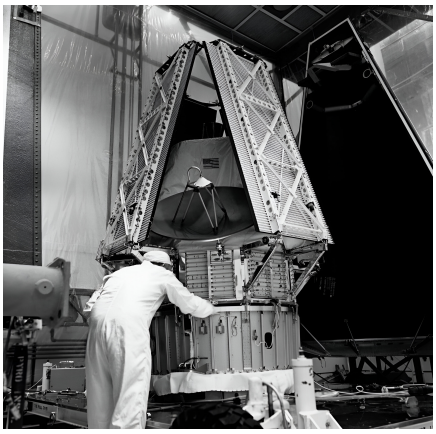
Left: Richard "Sam" Bogner (1931–2015), chemical engineer from Decatur, Indiana; battery system project engineer at NASA JPL for Surveyor and Mariner programs, Hughes Space and Communications, and GM SunRaycer. **Right:** The mid-century entrance to NASA's Jet Propulsion Laboratory in Pasadena, California.

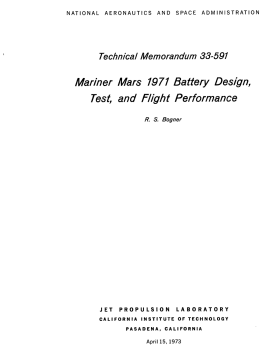


Surveyor 3 on the Lunar Surface (Oceanus Procellarum): Apollo 12 Commander Pete Conrad examines the Surveyor 3 lander in November 1969, with the Lunar Module *Intrepid* visible on the horizon. Sam Bogner qualified the silver-zinc battery packs that enabled Surveyor to survive launch vibration, lunar impact deceleration, and two-week cryogenic nights. (NASA Photo AS12-48-7135).

modeled the exact discharge curves required when Mariner 9 entered orbit and dipped into the dark shadow of Mars, and a bunch of other stuff I still don't fully understand. When Mariner 9 successfully outlasted a planet-wide dust storm to map 100% of the Martian surface, it did so powered by an architecture he engineered.

As the space race began to transition into the commercial satellite boom of the late 1970s and





NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

Technical Memorandum 33-591

Mariner Mars 1971 Battery Design, Test, and Flight Performance

R. S. Bogner

JET PROPULSION LABORATORY
CALIFORNIA INSTITUTE OF TECHNOLOGY
PASADENA, CALIFORNIA

April 15, 1973

Left: Technicians and engineers in cleanroom garments conduct final system checks on the Mariner 9 spacecraft prior to its May 1971 launch to Mars (NASA/JPL). **Right:** NASA Technical Memorandum JPL-TM-33-591 (April 1973), *"Mariner Mars 1971 Battery Design, Test, and Flight Performance"* authored by R. S. Bogner.

1980s, Uncle Sam brought his expertise to Hughes Space and Communications. At Hughes, he moved away from the volatile inner solar system and focused on longevity and weight reduction for Earth's orbital infrastructure. He pioneered research into lightweight, graphite-epoxy battery casings to replace heavy metallic housings, testing nickel-cadmium cells through simulated geosynchronous orbit-eclipse seasons to ensure communication satellites could survive for decades without failing.

This work ties directly into the next portion of his career, and takes him back to his beginnings in the automotive industry. In 1987, General Motors partnered with AeroVironment and Hughes to build the SunRaycer, a revolutionary solar-powered vehicle designed to contest the inaugural World Solar Challenge across the Australian Outback. On the SunRaycer project, he functioned as a primary battery design engineer and lead technical researcher representing Hughes Aircraft. He is officially credited in the Society of Automotive Engineers (SAE) archives as one of the definitive engineers of the vehicle's power architecture, co-authoring the official technical documentation titled: *"GM Sunraycer Case History Lecture 3-4: Design and Performance of the GM Sunraycer Battery"* alongside his colleagues N. Chorneau and D. Pickett. In this capacity, he helped engineer a custom, ultra-lightweight silver-zinc secondary battery pack. This pack had to flawlessly store transient energy harvested from the car's cutting-edge gallium-arsenide solar array, delivering bursts of raw power when navigating hills or cloudy stretches. And he unequivocally delivered. The engineering was so pristine that the GM SunRaycer didn't just win the 1,800-mile race across the continent. It crossed the finish line more than two full days ahead of the nearest competitor. The car currently sits in the Smithsonian in Washington, D.C.

I would love to say that I grew up listening to my great-uncle Sam tell stories about NASA, rocket ships, or the space race. Unfortunately, I don't recall ever meeting him. My grandpa Alan mentioned his older brother from time to time, but as a kid, I don't think I ever paid much attention. Then one day, less than a year ago, my dad casually mentioned his work on the SunRaycer. I was beyond intrigued. That was when I discovered all of this for the first time. So, while I was never regaled with stories of his work in space—which, even as a child, I would have found profoundly interesting—I hope the work that follows, enabled by his early research at JPL, might continue, in a much smaller way, my family's legacy on the Red Planet.



The GM SunRaycer on Museum Display: Preserved at the Smithsonian National Museum of American History in Washington, D.C., the pioneering solar vehicle was powered by custom silver-zinc secondary cells engineered by Sam Bogner and Hughes Aircraft, winning the 1,867-mile World Solar Challenge across the Australian Outback in 1987 by more than two days.

I didn't think I could ever do anything as interesting. It turns out that today, I could do a surprising amount of it. NASA publishes raw images and telemetry from its rovers, including pointing direction, focus distance, and solar angles. And modern AI, when rigorously guided by human oversight, is much better at math than I will ever be.

What I actually did: Dissecting the data

Before you can ask what something is, you have to find out what it actually measures. Here is what happened when we dug into NASA's raw files:

- **We busted the "20-centimeter" rumor.** Online commentators claimed the object was 20 centimeters long by assuming it was sitting on a distant cliff 45 meters away. But NASA's telemetry logs showed the camera's focus motor count was set to 2705, meaning the lens was focused at exactly **3.96 meters** from the rover. The object isn't on a distant cliff; it's sitting right in front of Curiosity. That dropped its true dimensions to roughly **3.5 centimeters long and 11 millimeters across** or about the diameter of a AA battery.
- **We tested the shadow with solar math.** The mission records showed the Martian Sun was sitting at an elevation of 69.8° (nearly straight overhead at noon). At that angle, an object's shadow on the ground must be compressed to just 37% of its physical height. When we measured the shadow fringe beneath the cylinder (9.8 ± 0.7 pixels) and compared it to what the Sun's position dictated, both optical scales—the normal transverse camera scale (0.293 mm/px, yielding 7.81 mm relief and a 10.76 mm full diameter) and the inclined ground slope scale (0.317 mm/px, yielding 8.46 mm relief and an 11.66 mm diameter)—bracket the

physical diameter to 11.2 ± 0.5 mm ($< 8.3\%$ spread). That proved beyond doubt that this is a real, rigid, three-dimensional physical object casting a genuine shadow on the soil, not a digital smudge, a compression defect, or an optical illusion.

- **We pulled the boundaries apart with edge filters.** To keep my eyes from tricking me, I had the AI write code applying spatial "Sobel gradient filters." This math measures how abruptly brightness changes across pixels. It proved that the dark circle on the end wasn't a shadow thrown on the ground, it was the cylinder's own flat, self-shadowed face, completely separate from the neighboring rock and the dirt beneath it.

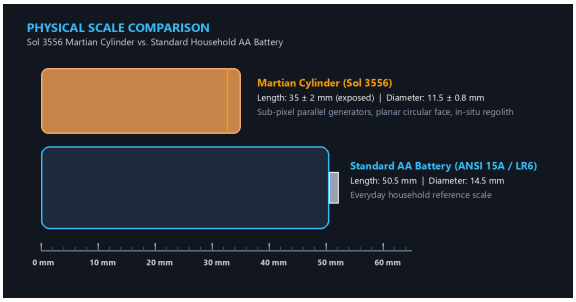


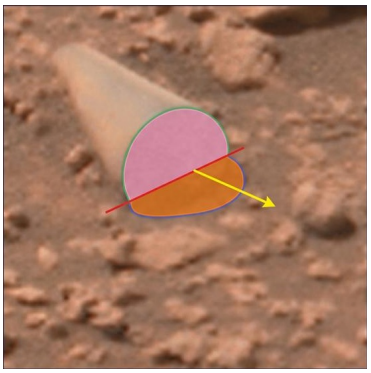
Resolving the 20-Centimeter Rumor: The full 1328×1184 pixel Mastcam-100 field of view (Sol 3556). Internet commentators assumed the cylinder was embedded in the distant Bolivar cliff face (~ 45 m away), which would scale a 35-pixel feature to ~ 20 cm. NASA's camera focus motor telemetry (`MSL:FOCUS_POSITION_COUNT = 2705`) proves the lens was locked onto the foreground regolith at 3.96 m, constraining the true exposed length to 35 mm.

Building the tool and checking the back edge

Ten years ago, I couldn't have done this analysis because the tools were out of reach. Reading NASA's data format, working out camera geometry, writing code for edge detection, fitting ellipses, running the error checks… each of those is a specialty.

AI collapsed that barrier. It wrote the Python code that reads the data format and calculates the center of the image. It explained what a "Sobel filter" is. It did the arithmetic I can't do and the geometry I never learned. When I asked "what does that mean?", it explained, as many times





Left (Physical Scale Comparison): With a diameter of 11.5 ± 0.8 mm and an exposed length of 35 ± 2 mm, the Martian cylinder is roughly the girth of a standard AA battery (14.5 mm diameter) and about two-thirds its length. **Right (Solar Ephemeris Ray-Tracing):** Decoupling the object's self-shadowed planar face (pink) from its ground cast shadow fringe (orange) under the 69.82° noon solar elevation.

as I needed.

> **The barrier to rigorous analysis is lower than it has ever been.**
>
> But here's the part people skip. **The AI got things wrong. Repeatedly.** It was honestly exhausting and at some points infuriating. The things that it couldn't seem to do, even with its ability to do calculus and trigonometry in seconds, were to me, dead simple.

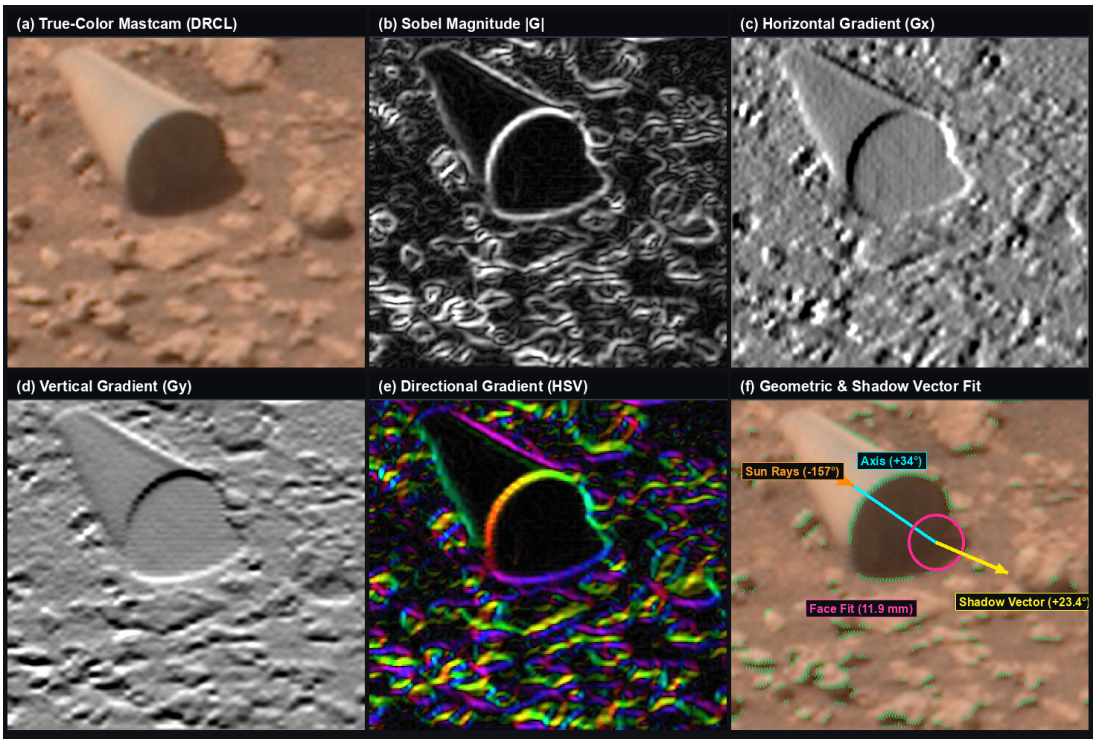
It kept making the same kinds of mistakes:

- **It mixed up two different shadows.** The cylinder's end face is dark because it's turned away from the Sun. That's *its own* shading. Separately, the cylinder throws a shadow on the ground. The software saw "dark" and lumped them together, then built measurements on a shape that didn't exist. I could see, just by looking, that those were two different things.
- **It tried to fit a perfect circle to something that wasn't one.** When you look at a circle from an angle, it becomes an ellipse. The AI kept forcing a circle onto a tilted view. It's the kind of mistake that most intelligent people wouldn't make, mostly because they have spent a lifetime of looking at shadows and shapes in the world.
- **It swallowed a neighboring rock.** Next to the cylinder was a lump that wasn't part of it. The software kept including it in the outline until I told it to leave that part out.

I also had the AI build a custom web-based photogrammetry tool that allowed me to overlay a 3D cylinder model directly onto the NASA photo and manipulate its rotation, perspective tilt, and length in real-time. Working with that 3D model alongside the Sobel edge filters helped clarify two things about the object's shape and its back end:

First, because a true cylinder has the same physical radius at both ends, locking the model to Curiosity's 22° downward camera angle confirmed that the object isn't a pointed cone or a tapered rock shard. At four meters away through a telephoto lens, optical perspective shrinks the back end by less than a sixth of a pixel (0.8%). That means an untapered cylinder *has* to show parallel sides all the way back—which is why the front and rear have the same physical radius to within measurement precision.

Second, people had assumed the cylinder was either a broken stub or a continuous rod plung-



Sub-Pixel Spatial Gradient Forensics: Six-panel computer-vision analysis showing raw crop, horizontal Sobel gradient (S_x) isolating the left circular rim, total gradient magnitude, direct Fitzgibbon ellipse fit ($b/a = 0.9335$) and parallel longitudinal generators ($R^2 = 0.9985$), directional gradient field, and transverse cross-sectional profile.

ing indefinitely into the Martian dirt. When we checked the back tip where the sunlit highlight meets the soil using the same Sobel edge filter we used on the front face, the upper half of that boundary didn't just fade out along the straight dirt line; it traced a crisp, curved rim turning through 112° of arc right where the barrel ends and meets the regolith. Even if a skeptic argues that the sloping dirt at the bottom is hiding more of the shaft, the exposed barrel forms a self-contained cylinder at least **110 pixels by 34 pixels**, giving a length-to-diameter aspect ratio of 3.24 : 1 (≈ 35 mm $\times$ 11 mm).

Notice what I contributed. It wasn't equations. It was **judgment**. I knew what a shadow looks like. I knew how a cylinder behaves. I knew when something looked wrong. The AI had the mathematical ability. I had the *common sense* to tell it when it was pointed in the wrong direction.

Embodied experience still matters

I think this is the most important thing I learned, and I'd like you to take it away from this article.

An AI has read about shadows. I have *stood* in them. It knows the word "perspective." I've driven through a tunnel and seen the end getting closer. It can describe a coin on a table. I've flipped one and watched the circle become an ellipse.

That lived, physical understanding of how things look and behave is not a small thing. It's what let me notice, quickly and without any formal training, when the analysis had drifted away from reality.

AI can be confidently, fluently wrong. It can produce a clean chart and a convincing explanation of a result that's broken.

The people who are going to excel moving forward won't be the ones who trust AI the most or the least. They'll be the ones who combine **powerful tools with real-world wisdom** and stay in charge of the process.

Why this actually matters

Why does measuring a tiny 3.5-centimeter object on Mars matter?

1. **It replaces wild speculation with checkable facts.** In an online landscape divided between wild claims ("it's proof of an alien probe!") and lazy dismissal ("it's just a rock, stop looking"), numbers provide sanity. Anyone with a computer and curiosity can download NASA's Sol 3556 dataset, run the equations, and verify the dimensions for themselves. That is how science is supposed to work.
2. **It shows that curiosity doesn't need credentials to be rigorous.** You don't need a PhD or a corporate lab grant to audit space agency data. The data belongs to the public. If you are willing to learn, ask hard questions, and refuse to accept comfortable assumptions, you can do genuine planetary forensics from your living room.
3. **It demonstrates the future of human-AI collaboration.** AI isn't an oracle that hands down absolute truth, nor is it a useless novelty. It's an exoskeleton for human curiosity. When guided by someone who refuses to let it hallucinate, it allows one person to do the work that once required an entire research team.

You can do this too

NASA's rover images are public. So are images from the Hubble and James Webb telescopes, satellite archives, weather data, seismic logs, and thousands of other datasets. For most of history, serious investigation required a university, a lab, and a funding source. Now it requires **curiosity, patience, and a willingness to be corrected.**

If you're thinking about trying something like this, here's what I'd suggest:

1. **Start with a question you can actually answer.** Not "is this alien?" but "how big is this, really?" Small, measurable questions lead somewhere.
2. **Use the original data, not screenshots.** Compressed social media images throw away exactly the information you need.
3. **Let AI do the heavy lifting, but never the thinking.** Ask it to explain everything. Ask it to show its work. Ask "how would I know if this were wrong?"
4. **Trust your eyes and your instincts.** If a result looks off, make the software justify itself.
5. **Be honest about what you don't know.** The most credible sentence in my paper is the one that says that I don't know what the object is.

6. **Share your work openly.** Let other people check it, argue with it, and break it. That's how science works.

What happens next

I still don't know what that object is. But now I know that it's actually cylindrical. And I know that it shouldn't be there.

In the technical paper, I evaluated the three competing hypotheses:

1. Terrestrial rover or landing debris

The most intuitive assumption is that Curiosity dropped a bolt, spacer, or piece of hardware, or that debris drifted from the 2012 Entry, Descent, and Landing (EDL) event. But this runs into severe physical problems:

- **The distance and supersonic wind gap:** Curiosity's landing site (Bradbury Landing), where the sky crane and heat shield crashed, is over 28 kilometers away along the rover's winding drive path, and roughly 13 to 15 kilometers (8 to 9 miles) in a direct straight line—plus a 610-meter climb uphill onto Mount Sharp. When *Perseverance* found a piece of thermal blanket in 2022, it was paper-thin foil that catches the breeze like a kite. In Mars's thin atmosphere (less than 1% of Earth's air density), lofting a solid AA-battery-sized metal cylinder requires **supersonic winds of 750 to 1,300 miles per hour (Mach 1.4 to Mach 2.4)**. Martian gusts in Gale Crater top out around 75 mph.
- **The rover's own diary says its hands were in its pockets:** What if Curiosity dropped a part right there? Inside the raw data file is a mechanical log called the Rover Motion Counter. At that stop, the counters for the drill, soil scoop, dust brush, and sample funnel were all sitting at zero. Even better, Curiosity's robotic arm sits on its front bumper and can only reach 2.1 meters—while this object was photographed nearly 4 meters away behind the rover's left shoulder, on untouched ground with no wheel tracks in sight.
- **It doesn't match the spare parts list:** We checked NASA's engineering blueprints for every cylindrical part on the rover. Curiosity's drill bits are 16 millimeters wide (nearly 50% fatter) and carved with deep spiral grooves like a masonry bit. Aerospace bolts have wide hexagonal heads, external threads, and safety wire. Pin-pullers and wax actuators keep their pins locked inside housings so they can't litter the planet. And the pieces of broken wheel tread Curiosity has shed over the years are flat, 1.5-millimeter-thick strips of aluminum, not solid round rods.
- **The sediment clock (80 to 320 years) and why it couldn't sink:** The cylinder isn't resting loosely on top of fresh dust. Its lower third (≈ 2 to 3.2 mm of depth) is embedded in the crusted soil matrix along a straight ground interface. Could a heavy metal part have simply sunk into soft dirt under its own weight? Under Mars's weak gravity, a AA-battery-sized cylinder of aluminum or steel weighs less than a few coins and presses down with less than 0.3 kilopascals—while Curiosity's wheel sensors proved Gale Crater's salt-crusted dirt is **50 to 500 times too hard** for it to sink in. Nor could a passing windstorm have banked sand around it overnight: wind blowing past an obstacle carves a horseshoe-shaped trench ("scour moat") in front and leaves a tapered sand tail behind it, and neither is there. In Gale

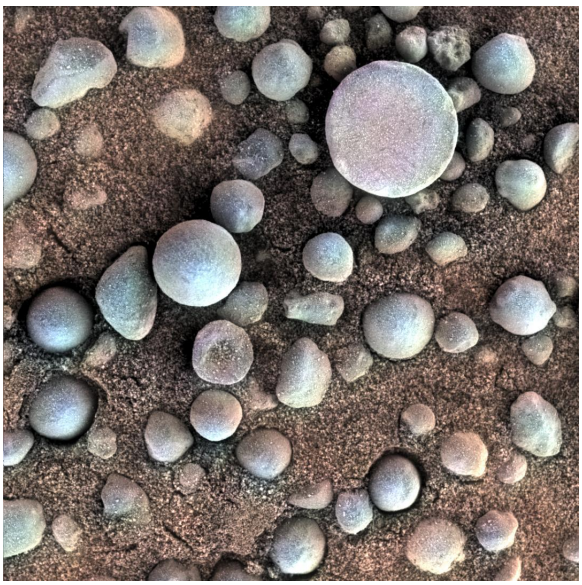
Crater, passive atmospheric dust settles at only 10 to 25 μm per Earth year (and rover UV sensors show seasonal winds scrub much of that away). Accumulating 2 to 3.2 millimeters of natural sediment fallout requires roughly **80 to 320 years** (80 to 200 years for a conservative 2-millimeter vertical depth). Curiosity had only been on Mars for 10 years at the time of the photo, and no other mission in history has ever landed within 500 kilometers of Gale Crater.

- **No mechanical trauma:** The cylinder shows no sheared mounting brackets, severed cabling, or buckling from a high-velocity ballistic impact ($R^2 = 0.9985$ straightness).
- **Tested against surrounding rocks:** When we verified an empirical control sample of twenty other rocks across the same frame (extracting each rock directly from the raw pixel detector), natural basalt and sandstone clasts averaged an aspect ratio of 1.23 and a generator straightness fit of only 0.10, with none scoring above 0.271. The cylinder's 0.9985 straightness is an extreme ten-sigma statistical outlier (+10.70σ, Cantelli bound $p < 0.009$).

2. A wind-carved rock (ventifact) or mineral concretion

The standard geological dismissal is that it is just a rock shaped by natural forces. Better yet, we don't have to guess what Martian ventifacts or mineral concretions look like—we can compare the cylinder directly against NASA's own textbook photos of them:





Left (What Real Gale Crater Ventifacts Look Like): *Curiosity* Left Navcam image (`NLB_700661467EDR_F0933078NCAM00299M_`) taken on Sol 3415 atop the nearby Greenheugh pediment, featured in NASA's *"Sols 3417–3418: A View Filled With Ventifacts"* mission update. Wind-blown sand carves flattened facets and sharp angular keels (*Dreikanter*), not smooth round cylinders (NASA/JPL-Caltech). **Right (What Real Martian Concretions Look Like):** *Opportunity* Microscopic Imager view of hematite-rich mineral concretions ("blueberries") near Fram Crater (3 cm across, Sol 84, NASA Photojournal PIA19113). Groundwater precipitation grows spherical nodules ($L/W \approx 1.0$), not elongated straight-walled cylinders (NASA/JPL-Caltech/Cornell/USGS).

- **What is a ventifact—and what does NASA say they look like in Gale Crater?** A ventifact (from the Latin for "wind-made") is a rock that has been sculpted, faceted, or polished by wind-blown sand over thousands or millions of years. Think of natural sandblasting. Just

$\sim$ 140 sols before taking the cylinder photo, *Curiosity* drove onto the nearby Greenheugh pediment and found itself surrounded by hundreds of genuine Gale Crater ventifacts (above left). In NASA's official mission blog (*"Sols 3417–3418: A View Filled With Ventifacts"*), JPL Planetary Geologist Dr. Abigail Fraeman defined them in one sentence: *"rocks which have been abraded by wind-blown sand into distinctive, flattened facets with crisp edges."* Look at those rocks in NASA's Sol 3415 photo: because wind blows from dominant directions, saltating sand carves flat slanted faces, sharp triangular ridges (keels), and jagged wedges—not smooth, round cylinders.

- **Wind is not a lathe:** While natural sandblasting easily produces faceted stones or rounded pebbles (several natural rocks in the surrounding field score over 90% in projected roundness, meaning circularity alone is not an anomaly), it **does not act like a lathe**. If a rock is anchored in the dirt, wind carves flat angled facets and sharp ridges like the Greenheugh pediment rocks; if a pebble tumbles along the ground, its corners chip away first, rounding the ends into a football or egg shape and destroying any flat 90° circular face. The true anomaly is the non-parametric conjunction of three features that never appear together in natural geology:

  1. Two strictly straight, mutually parallel sides ($R^2 = 0.9985$ straightness, an extreme ten-sigma anomaly of $+10.70\sigma$, Cantelli bound $p < 0.009$, compared to the flattest natural rock in the scene at $R^2 = 0.2708$).
  2. A crisp, flat planar end cap terminating perpendicular to the body, whose 0.9335 apparent circularity perfectly matches the 22° camera perspective.
  3. An elongated 3.24 : 1 length-to-width aspect ratio (over twelve standard deviations, $+12.06\sigma$, Cantelli bound $p < 0.007$, away from the roughly equal dimensions of surrounding gravel).

- **Concretions grow into spheres or crusty hollow pipes, not machined rods:** As shown in NASA's famous close-up of Martian "blueberries" near Fram Crater (PIA19113, above right), mineral nodules formed by chemical precipitation in groundwater grow outward equally in all directions, creating spheres ($L/W \approx 1.0$) or lumpy clusters. Even on Earth, in places like Utah's Navajo Sandstone where groundwater cements iron oxide into "pipe concretions," those pipes have bumpy, flaring sandstone crusts and hollow centers—more like a crusty piece of rigatoni than a smooth, solid cylinder with a flat circular face. And natural six-sided mineral crystals have flat polygon sides that gleam in discrete steps, whereas our Sobel directional filter proved the cylinder curves smoothly all the way around.

3. An artifact of unknown provenance

Because the cylinder displays manufactured tolerances, yet cannot be easily explained by drifting landing debris or natural geology, scientific rigor demands keeping a third possibility on the table: that it could be an artificial artifact of unconfirmed origin.

When I say "something stranger," I mean something that most mainstream planetary science is hesitant to say out loud, but that every curious person wonders about in the quiet of their thoughts: could Mars have once hosted an ancient technological presence? If I am being completely candid about what drives my imagination, the spark of curiosity that kept me up late staring at pixel grids, I sincerely hope that this cylinder might turn out to be evidence of that.

Theoretical plasma physicist Dr. John Brandenburg pointed to isotopic Xenon-129 anomalies on Mars. Penn State astrophysicist Dr. Jason Wright published papers detailing how durable non-biological artifacts (technosignatures) could survive buried in planetary soil for billions of years. When I look at this clean cylinder with its flat circular end face, I think about those models.

Do I know that's what it is? Absolutely not.

And that honesty is what keeps this work credible. The math does not prove an ancient artifact; it only proves that the object is small, physically real, precisely shaped (3.5 cm × 1.1 cm), and unaccounted for by conventional optical glitches or known rover shedding.

The honest answer is that the analysis narrows the possibilities, and a close-up look from Curiosity's contact instruments, including the MAHLI hand lens (resolving microscopic machining marks versus mineral crystal cleavage) and the ChemCam laser spectrometer (detecting alloy metals versus basaltic silicates), would do far more than any amount of image processing from Earth. That's what Avi Loeb asked for, and I agree with him.

The point isn't to jump to conclusions. The point is that *we are allowed to look, and we are allowed to wonder.*

Whatever the answer turns out to be, something has already changed. A person with no engineering degree and no math training was able to sit down with public data and produce a careful, checkable analysis.

The world is full of data. The tools are widely available. What it still needs is people willing to look carefully, ask honest questions, and use their own eyes and judgment to steer.

How This Was Built

I'm not an engineer or a mathematician. I'm a regular guy who loves rigorous inquiry into anomalous data. I used AI to write the Python scripts, parse NASA's PDS telemetry headers, run the linear algebra, and typeset the LaTeX paper. I told the AI what to test and caught the software when it did something counter-intuitive, like including the cast shadow into the end face of the object, or lumping in the rock next to it.

It didn't take months of grant funding or an institutional research team. It took an inquisitive amateur in Indiana, a few days, raw public NASA files, and an AI that can calculate without emotion.

For years, podcasts, social media, and some academics repeated claims that this was an eight-inch pipe, arguing over rover debris and alien artifacts without anyone actually opening the mission records to check. The telemetry was sitting on NASA's servers the entire time. You don't have to trust me, and you don't have to trust the AI. I didn't. The raw NASA files and `reproduce_analysis.py` are in the open GitHub repository and permanently archived on Zenodo (doi:10.5281/zenodo.23789092). Run the math yourself.

The accompanying technical paper, "Morphological and Photometric Analysis of an Unidentified Cylindrical Feature in Curiosity Sol 3556 Mastcam Imagery," contains the complete methods, equations, and figures for anyone who wants to check the work, or improve on it.